TVR-1

Mechanical Team - Project Kickoff

Overview

Parameter	Value	Parameter	Value
Airframe OD / ID	79.25mm / 76.2mm (3in)	Motor	AeroTechG61W, 38mm
Tube material	Wildman G12 fiberglass	Motor case	RMS 38/120, 106.7mmlong
Total length	94.4 cm	Stability margin	1.23 cal
Total mass (w/ motor)	~968 g	CG / CP (from nose)	55 cm / 64.4 cm
Flight computer	150 g (electronics only, excl. battery/motor/gimbal)	Gimbal max deflection	5°

Three main pieces

System	Scope	Status	Priority
Gimbal	Refine the existing gimbal design to flight-ready	Draft 1 complete	Primary
Electronics Bay	Onboard mounting for the flight computer, battery, and wiring	Not started	Primary
Ground Station Enclosure	Handheld case for the USB-dongle-style ground station device	Not started	Secondary

(“Primary” = nothing flies without this. “Secondary” = important but not flight-blocking)

Gimbal

What exists already:

- Draft 1 modeled in Onshape - pivot geometry, bore, and mounting holes proven out.
- CG-to-nozzle lever arm: 0.394m (used for the rocket-level corrective torque calc).
- Bore sized at ~42.5mm with 3mm wall (48.5mm outer) to clear the tube with margin.
- Pivot pins: M3 socket head cap screws with nylon-insert locknuts.
- Physical deflection limit: 5° (this is what the control-loop sim is built around)

What "improve" means:

- Tighten tolerances so the pivot moves smoothly without excess play.
- Cut unnecessary mass where the structure allows it.
- Make it serviceable, should be possible to inspect/replace the pivot hardware between flights without rebuilding the assembly.
- Nail down the servo-to-pushrod lever arm (separate from the CG-to-nozzle lever arm above), this determines the real servo torque spec, and it's currently unknown until the linkage geometry is finalized. This is the single most important open item.

Electronics Bay

What it needs to do:

- Physically mount the flight computer board (Samantha's design), battery, and wiring harness inside the airframe.
- Stay within the 76.2mm clear bore.
- Keep the bay's mass centered near the 55cm CG point, don't let this section quietly shift the balance.
- Provide an access panel or removable sled so boards can be swapped or inspected between flights without disassembling the whole rocket.

Open items:

- Final board dimensions from Samantha (will update once she has dimensions for her board size, she is still in schematic process)
- Battery size/chemistry, will confirm with Samantha locking mounting geometry.

Ground Station Enclosure

What it is

it's the handheld ground station device Yusuf is building (currently in schematic phase). Plugs into a laptop, receives telemetry over LoRa.

What it needs to fit:

- A small PCB with: LoRa module, USB-UART bridge, voltage regulator, status LED, and associated passives.
- An exposed USB-A connector on one end.
- Compact, pocketable form factor, think actual USB drive size, not a project box.

Open items:

- Final PCB outline from Yusuf (will update once the layout progresses) before finalizing the enclosure's internal dimensions.